

Father's Day Crossword

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1	2
3	

ONE ACROSS

Dad cuts three positive real ribbon lengths a, b, c for wrapping three Father's Day gifts. They satisfy

$$a + b + c = 14$$

$$(a - b)^2 + (b - c)^2 + (c - a)^2 = 32$$

$$(a + b - c)(b + c - a)(c + a - b) = 104.$$

Find the value of abc .

ONE DOWN

Dad designs a Father's Day lock whose possible codes are residue classes modulo

$$M = 3 \cdot 5 \cdot 7 \cdot 11 \cdot 13 \cdot 17.$$

A code x is called *reversible* if

$$x^2 \equiv 1 \pmod{M}.$$

How many reversible codes are there, excluding the code $x \equiv 1 \pmod{M}$?

TWO DOWN

Ten gift boxes are placed at equally spaced points around a circular table. Dad wants to connect the boxes in pairs using five ribbons, with every box connected to exactly one other box.

The ribbons must be drawn inside the circle, and no two ribbons may cross.

How many different ribbon-pairing patterns are possible?

THREE ACROSS

In Dad's backyard, two circular grills have centers A and C . The first circle has radius 20, the second circle has radius 34, and the distance between their centers is $AC = 42$. The two circles intersect at points P and Q .

Find the exact length of the common chord PQ .